RESEARCH ARTICLE

An Improved Transformer-Based Model for Urban Pedestrian Detection

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Abstract
Pedestrian detection is a crucial task in computer vision, applicable in object tracking, video surveillance, and autonomous driving. Recent years have witnessed substantial advancements in pedestrian detection due to the fast evolution of deep learning in object detection. Nonetheless, obstacles such as inadequate detection accuracy persist, mostly because of varied pedestrian postures and intricate environments. This study proposes the RT-DETR-improved model to overcome these issues based on the real-time detection transformer (RT-DETR). First, we incorporate the high-low frequency (HiLo) attention into the encoder, therefore enhancing the model's detection performance. Furthermore, we present a nonlinear feature fusion module that fuses information from various feature scales and contexts more successfully. We also introduce a novel loss function, InnerMPDIoU, to enhance detection efficacy in congested environments. To evaluate our model's performance, extensive experiments are conducted on the CityPersons dataset. Compared to the baseline model, the RT-DETR-improved model attains a 4.2% enhancement in mAP50, a 2.0% improvement in mAP, a 2.2% rise in accuracy, and a 3.1% gain in recall. The results demonstrate that the proposed method exhibits superior detection accuracy and robustness.

Keywords CityPersons dataset · HiLo attention mechanism · InnerMPDIoU · Nonlinear feature fusion · Pedestrian detection

1 Introduction

Pedestrian detection is an essential task in computer vision, focused on recognizing and locating pedestrians inside pictures or videos. It is integral to applications including autonomous driving [1], pedestrian re-identification [2], pedestrian tracking [3], and traffic surveillance [4]. Earlier pedestrian detection methods predominantly depended on manually designed features. In 2005, Dalal et al. [5] proposed an algorithm employing histograms of oriented gradients (HOG) for feature extraction and a linear support vector machine (SVM) for classification. This method achieved an optimal equilibrium between rapidity and precision, establishing a standard for forthcoming pedestrian detection algorithms. In 2009, Dollar et al. [6] introduced the integral channel feature (ICF) to overcome the shortcomings of HOG, therefore improving both speed and accuracy in pedestrian detection.

While hand-crafted features may be successful in uncomplicated scenarios, they exhibit limited generalization capacity and fail to adequately capture high-level semantic information. Due to the robust feature representation capabilities of convolutional neural networks (CNNs) [7], CNN-based detectors [8–13] have attained considerable success in generic object detection tasks. As a result, pedestrian detection has progressively transitioned from dependence on manually crafted features to features derived from deep learning. The detection accuracy of CNN-

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based algorithms has been significantly enhanced in comparison to traditional methods. Nonetheless, obstacles persist, especially in identifying tiny things and handling intricate environments. The newly introduced end-to-end detection transformer (DETR) [14] has exhibited performance that is equivalent to or even exceeds that of traditional object detection methods. DETR is proficient at obtaining long-range dependencies and global contextual information. Additionally, it utilizes a bipartite matching method to correlate queries with the ground truth, hence preventing repeated proposals. These attributes indicate that DETR possesses considerable promise for improving pedestrian detection by overcoming the constraints of CNN-based approaches. We propose an enhanced pedestrian detection model based on the RT-DETR [15], called RT-DETR-improved. The main contributions of this paper are as follows:

- To improve detection performance while maintaining low complexity, we introduced the HiLo attention mechanism. By dividing high- and low-frequency features, it ensures that different types of features receive appropriate attention.
- To augment the model's robustness and expressiveness in intricate scenarios, we present a nonlinear feature fusion module that boosts feature interaction nonlinearity, enabling the model to more efficiently assimilate information from diverse feature scales and contexts.
- To enhance detection performance in crowded environments, we introduce a new loss function, InnerMPDIoU, which adaptively modifies the auxiliary bounding box dimensions according to sample scale and streamlines the similarity assessment of bounding boxes.

2 Related Work

2.1 CNN-Based Pedestrian Detection

In light of the success of CNN-based detectors in generic object detection tasks, many CNN-based pedestrian detection methods [16–20] have been introduced, generally classified into one-stage and two-stage methods. Initially, two-stage detectors predominated in the application of CNNs to object detection tasks. In 2017, Li et al. [21] introduced the scale-aware Fast R-CNN (SAF R-CNN) framework, derived from Fast R-CNN [22], including two distinct sub-networks: one for identifying large pedestrians and another for small pedestrians. The model effectively identifies pedestrians of different sizes by employing scale-aware weights acquired during training. The integration of the two sub-networks enhance detection accuracy and speeds up the whole detection process. In 2019, Brazil et al. [23] proposed an autoregressive pedestrian detection framework (AR-Ped). The architecture presents an innovative stacking decoder module with convolutional re-sampling layers to enhance features across various sizes and layers. The autoregressive characteristics of AR-Ped allow the model to progressively enhance spatial features and classification predictions, resulting in increased accuracy.

A number of one-stage detector variations have been effectively utilized for pedestrian detection. Wei Liu et al. [24] introduced the ALFNet pedestrian detection method, which has an asymptotic localization fitting (ALF) module. This module employs a set of predictors to enhance the default anchor boxes of SSD [9]. Throughout the anchor box generation process, the IoU threshold is incrementally elevated, yielding an increasing quantity and superior quality of anchor boxes in proximity to the target, hence enhancing detection accuracy. In 2019, Liu et al. [25] presented the center and scale prediction (CSP) detector, an innovative method that recharacterized pedestrian detection as a high-level semantic feature detection task. The CSP detector has an anchor-free design, hence obviating the necessity for intricate post-processing techniques, which enhances the efficiency and simplicity of the detection process. The YOLOv8-CB [26] method was presented to enhance detection performance in densely populated pedestrian environments. This method integrates a lightweight cascade fusion network (CFNet) and a convolutional block attention module (CBAM), resulting in enhanced accuracy and reliability in pedestrian identification.

2.2 Transformer-Based Pedestrian Detection

Transformer-based end-to-end detectors (DETRs) have demonstrated outstanding performance in object detection and have been effectively utilized in pedestrian detection applications. In 2020, Lin et al. [27] presented the pedestrian end-to-end detector (PED), effectively utilizing the DETR model for pedestrian detection. The primary contribution of this model is the development of an innovative DETR decoder method, dense queries and rectified attention field (DQRF), which improves fine localization and feature correlation of identified objects. PED implements a data augmentation method utilizing visible region labeling to mitigate the prevalent occlusion issue in pedestrian detection, hence enhancing the model's robustness in intricate occlusion situations. Furthermore, PED enhances the traditional dichotomous matching algorithm by integrating domain-specific prior information, markedly expediting the matching process and increasing efficiency by as much as 10 times, hence substantially augmenting the DETR framework's performance in pedestrian detection tasks. In 2021, Irtiza et al. [28] evaluated the generalization capabilities of convolutional neural networks and transformer networks with an identical detection architecture. Although transformers excel in some dataset assessments, CNNs provide superior generalization capabilities compared to transformer networks. Nevertheless, transformer networks remain sort of developing in relation to CNNs. As transformers undergo further optimization, the forthcoming generation of pedestrian detection frameworks may transition from CNNs to transformers. In 2022, Yuan et al. [29] proposed a spatial and multi-scale feature enhancement module based on a single encoder of the deformable vision transformer. Through experiments, they demonstrated its effectiveness in pedestrian detection. This module achieved an optimal balance between speed and accuracy.

To address the challenges posed by complex environments, researchers have made numerous efforts. In 2023, to improve the accuracy of pedestrian occlusion detection, Ma et al. [30] proposed a fuzzy pedestrian detection method based on DETR. The model introduced dynamic deformable convolutions to enhance feature representation capabilities. Additionally, a cascaded transformer decoder was employed to generate high-quality queries, reducing the impact of low-quality queries on detection performance during training and thereby improving detection accuracy. To enhance the performance of all-weather pedestrian detection, Xing et al. [31] proposed a multispectral pedestrian detection transformer (MS-DETR) in 2024. This model introduced a multi-modal transformer decoder to fuse visible and thermal features. Additionally, a loosely coupled fusion strategy was proposed to strengthen feature fusion.

3 Method

3.1 The RT-DETR Algorithm

The RT-DETR is the first end-to-end real-time object detector. A key innovation of this algorithm is its efficient hybrid encoder structure. By decoupling intra-scale feature interaction from cross-scale information fusion, it efficiently processes multi-scale features, significantly boosting computational speed while maintaining detection accuracy. The hybrid encoder comprises an attention-based intra-scale feature interaction module (AIFI) and a CNN-based cross-scale feature fusion module (CCFM). The AIFI module uses an attention mechanism to enhance high-level features of the backbone, identify relationships between conceptual entities, and improve object localization and recognition. This module enables the model to focus on critical regions, reducing computational demands and accelerating processing without sacrificing accuracy. The CCFM module integrates features across scales, enabling effective detection of objects of varying sizes and improving performance in complex scenarios. RT-DETR introduces an IoU-sensitive query selection mechanism to enhance the model's ability to capture target locations by improving query initialization quality. Specifically, this strategy dynamically optimizes the generation of query vectors based on the IoU between the candidate box and the ground truth, better aligning with the geometric characteristics of the target and improving the detection accuracy. Finally, RT-DETR flexibly controls inference

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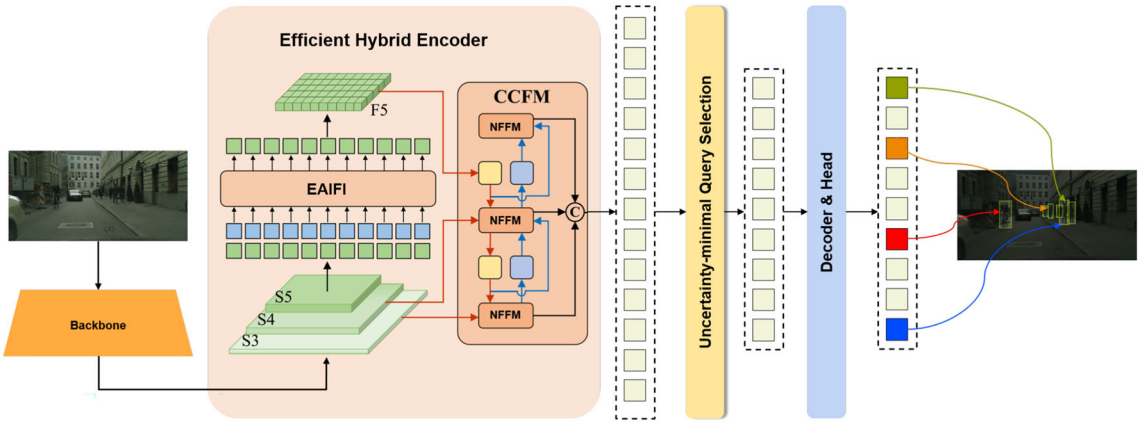


Fig. 1 The architecture of the RT-DETR-improved network

speed by adjusting the number of decoder layers, enabling a trade-off between speed and accuracy without retraining the model. This significantly enhances the adaptability and practicality of real-time object detectors in real-world applications.

In pedestrian detection tasks, both accuracy and real-time performance are essential. Leveraging the global modeling capabilities of Transformers, RT-DETR can effectively capture object dependencies in complex scenarios and maintain high detection accuracy, even with occlusion, complex backgrounds, or small objects. Compared to traditional DETR, RT-DETR enhances the model's inference speed by optimizing computational efficiency. Therefore, this study focuses on the latest end-to-end algorithm, RT-DETR, as the primary subject of our improvement research.

3.2 The RT-DETR-Improved Algorithm

3.2.1 The Overall Framework of RT-DETR-Improved

In this part, we present the RT-DETR-improved algorithm to enhance the overall performance of RT-DETR in pedestrian detection. The comprehensive architecture of RT-DETR-improved is introduced. The improvements to RT-DETR are described, encompassing the improved AIFI module, the nonlinear feature fusion module, and the InnerMPDIoU bounding box loss function. The network architecture of RT-DETR-improved is illustrated in Fig. 1.

3.2.2 Improvement of the AIFI Module

The AIFI module is similar in design to traditional transformer encoders, relying heavily on multi-head self-attention (MSA) mechanism. The MSA processes input in parallel and captures information from different representational subspaces, allowing each head to independently compute attention. This enhances the model's parallelism and expressiveness, enabling it to more effectively capture diverse dependencies in the input data. However, standard MSA layers often enforce uniform global attention across all image patches, overlooking the fact that features have varying fundamental frequencies. This limitation leads to substantial computational resource consumption when processing high-resolution images. To address this issue, we an enhanced AIFI (EAIFI) module by introducing the HiLo attention mechanism [32].

The HiLo attention mechanism processes high-frequency and low-frequency components of feature maps separately. It divides the same number of heads in the multi-head self-attention layer into two groups, with a ratio of α. Specifically, $(1-\alpha)N_h$ heads are allocated for high-frequency (Hi-Fi) processing, while αN_h heads handle low-frequency (Lo-Fi) processing, as illustrated in Fig. 2.

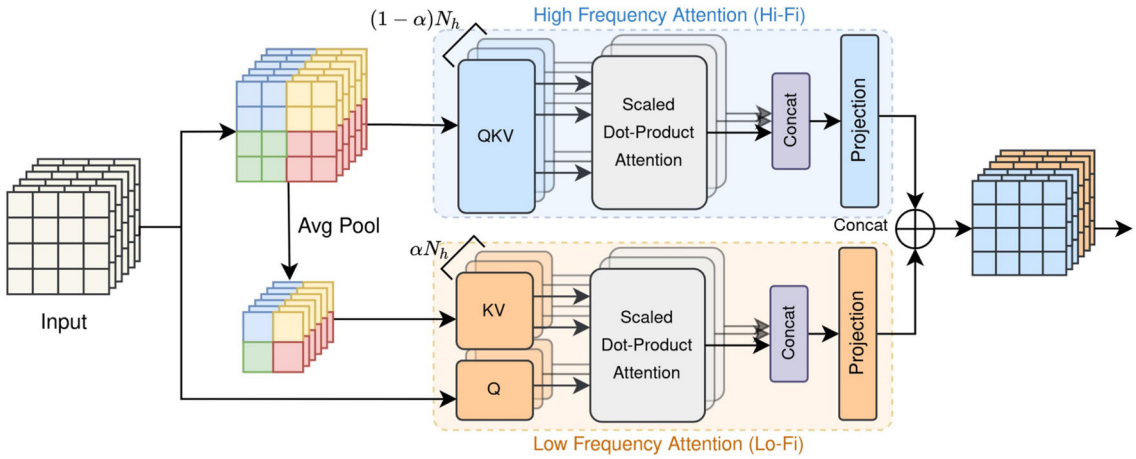


Fig. 2 The architecture of the HiLo attention

Specifically, HiLo divides the heads in the MSA into two groups. The Hi-Fi attention focuses on high-frequency details in the image, such as edges and textures, by capturing fine-grained high-frequency information through local window-based self-attention. Non-overlapping window partitioning is employed to reduce the range of self-attention calculations and redundant computations, significantly decreasing computational complexity. The Lo-Fi attention component processes low-frequency features obtained through average pooling within each window. After pooling, the feature map is projected into K and V, while Q is retained from the original feature map. Global attention is then applied to capture rich low-frequency information, such as shapes and color distributions, primarily operating on downsampled feature maps to reduce the number of features processed and improve efficiency. Finally, the high-frequency and low-frequency features are integrated to produce the output of the EAIFI module.

The MSA module in the original AIFI module is replaced with the HiLo attention module, as illustrated in Fig. 3. This improved structure fully leverages the high-frequency and low-frequency features of the image, reducing computational complexity while enhancing detection performance.

3.2.3 The Nonlinear Feature Fusion Module

The RT-DETR algorithm's feature fusion method has difficulties in capturing complicated feature interactions in intricate scenarios, resulting in erroneous or missed detections. The problem originates from the architecture of the fusion block in the CCFM module. The fusion block comprises N RepBlock units constructed from convolutional layers. The fusion method employs element-wise addition of outputs from the dual routes, which inadequately captures the semantic information in the input features. Motivated by [33], we introduce a nonlinear feature fusion module (NFFM) to augment the model's expressive capacity. The structure of the NFFM is illustrated in Fig. 4.

The NFFM substitutes the original RepBlock with the diverse branch block (DBB) and employs four separate pathways for feature fusion. The four DBB branches, from left to right, are obtained through multi-scale convolution transformation, convolution sequence merging, average pooling transformation, and Conv-BN merging. The multi-scale convolution transformation converts the $K \times K$ convolution layers in the DBB into standard convolution layers, enhancing the model's ability to perceive features at multiple scales. Convolution sequence merging combines the pointwise convolution layer in the DBB structure with the preceding convolution layer into one unified convolution layer. These transformations enhance the model's expressive power, enabling it to capture high-level semantic information in the input features, thus improving pedestrian detection accuracy. Mean pooling conversion transforms the $K \times 1$ convolutional layer in the DBB into a standard convolutional layer. Conv-BN combines the convolution operation with the batch normalization layer into a single operation, effectively reducing the model's computational load and accelerating inference speed. These structural changes allow DBB to intro-

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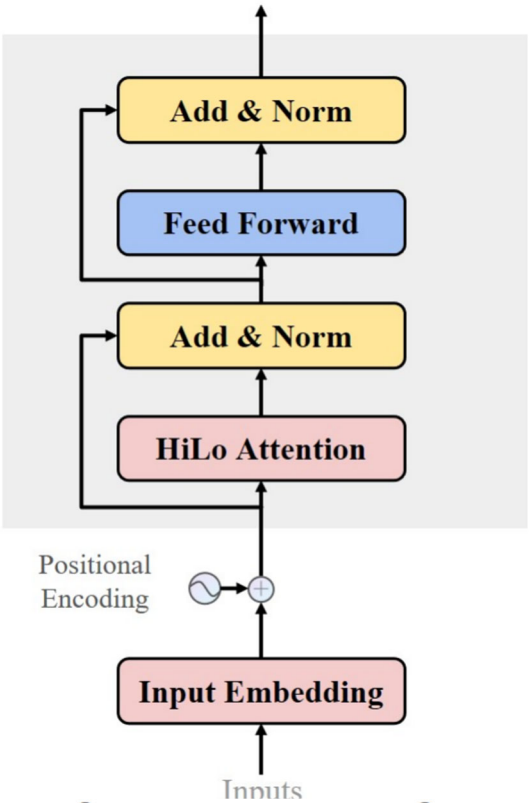


Fig. 3 The architecture of the EAIFI module

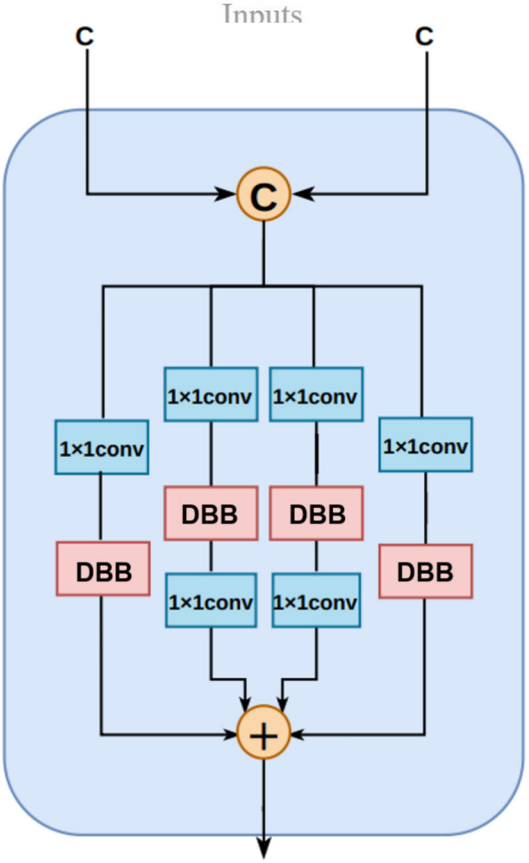


Fig. 4 The diagram of the nonlinear feature fusion module

duce additional complexity during training to optimize model performance, while during inference, it can revert to the original or simplified structure for efficiency. Overall, by combining DBB modules in various transformation forms, we can increase the number of paths with varying complexity and introduce more nonlinear operations, thereby improving the model's ability to detect pedestrians in complex backgrounds like occlusions.

3.2.4 The InnerMPDIoU

The bounding box regression (BBR) loss function is essential in pedestrian detection, significantly influencing model performance through its design. The sizes of pedestrians frequently differ in images, requiring bounding box designs that can effectively accommodate these discrepancies. Ineffective designs can lead to inconsistent predictions for pedestrians of varying sizes, therefore impairing detection performance. Moreover, the comparison of predicted boxes with ground truth boxes is particularly crucial in crowded scenes. An unsuitable similarity comparison approach may induce training instability and oscillations, adversely impacting the model's performance.

The IoU-aware query selection loss function in RT-DETR implements a query selection method that determines the highest IoU value between the predicted box and the ground truth boxes, utilizing this as the reference bounding box for subsequent loss computations. Nonetheless, it fails to consider multi-scale changes, leading to inconsistent pedestrian detection across different sizes. The loss function incorporates dynamic bounding box matching, enabling the selection of the most appropriate ground truth box for each predicted box. However, its performance diminishes in highly congested environments. This work introduces a novel loss function, InnerMPDIoU, to address these restrictions. The formulas for InnerMPDIoU are

$$b_l^{gt} = x_c^{gt} - \frac{w^{gt} * \text{ ratio}}{2}, b_r^{gt} = x_c^{gt} + \frac{w^{gt} * \text{ ratio}}{2} \tag{1}$$

$$b_t^{gt} = y_c^{gt} - \frac{h^{gt} * \text{ ratio}}{2}, b_b^{gt} = y_c^{gt} + \frac{h^{gt} * \text{ ratio}}{2} \tag{2}$$

$$b_l = x_c - \frac{w * \text{ ratio}}{2}, b_r = x_c + \frac{w * \text{ ratio}}{2} \tag{3}$$

$$b_t = y_c - \frac{h * \text{ ratio}}{2}, b_b = y_c + \frac{h * \text{ ratio}}{2} \tag{4}$$

$$\text{inter} = \left(\min\left(b_r^{gt}, b_r\right) - \max\left(b_l^{gt}, b_l\right)\right) * \left(\min\left(b_b^{gt}, b_b\right) - \max\left(b_t^{gt}, b_t\right)\right) \tag{5}$$

$$\text{union} = \left(w^{gt} * h^{gt}\right) * (\text{ ratio})^2 + (w * h) * (\text{ ratio})^2 - \text{ inter} \tag{6}$$

$$IoU^{\text{inner}} = \frac{\text{inter}}{\text{union}} \tag{7}$$

$$MPDIoU = \frac{IoU}{1 + d^2} \tag{8}$$

$$d = \sqrt{\left(x_c^{\text{gt}} - x_c^{\text{prd}}\right)^2 + \left(y_c^{\text{gt}} - y_c^{\text{prd}}\right)^2} \tag{9}$$

$$L_{\text{InnerMPDIoU}} = L_{MPDIoU} + \text{IoU} - IoU^{\text{Inner}} = \frac{IoU}{1 + d^2} + IoU - \frac{\text{inter}}{\text{union}} \tag{10}$$

where (x_c^{gt}, y_c^{gt}) denotes the center point of the ground truth (GT) box and the inner GT box, whereas (x_c and y_c) signifies the center point of the predicted box and the inner predicted box. The dimensions of the GT box are represented as x^{gt} for width and h^{gt} for height, while the width and height of the predicted box are represented by w and h. The variable $ratio$ represents the scaling factor, often within the interval [0.5, 1.5]. d denotes the distance between the center point of the GT box and predicted box.

The InnerMPDIoU integrates the features of the Inner-IoU [34], enabling it to calculate loss utilizing auxiliary bounding boxes of diverse sizes according to the sample scale. This allows InnerMPDIoU to more efficiently

Table 1 The overview of CityPersons dataset

	Train	Val	Test	Sum
Cities	18	3	6	27
Images	2975	500	1575	5050
Persons	19,654	3938	11,424	35,016

adapt to pedestrian targets of varying sizes, enhancing detection accuracy in multi-scale environments. Moreover, the InnerMPDIoU builds on the MPDIoU [35] concept, streamlining the similarity evaluation between bounding boxes, hence improving the efficiency of matching predicted and GT boxes. This reduces oscillations and instability during the training process. The proposed InnerMPDIoU, as a loss function that integrates bounding box scale adaptation and similarity simplification, effectively tackles obstacles such as multi-scale and dense environments in pedestrian detection, therefore improving overall detection performance.

4 Experiments and Analysis

4.1 Dataset

The CityPersons [36] dataset is a prominent standard for pedestrian detection, especially in urban settings. The dataset has 5050 photographs from 27 locations, depicting a variety of street scenes across different seasons and weather situations. Each photograph typically has around seven pedestrians. The dataset is divided into three subsets: 2975 images for training, 500 for validation, and 1575 for testing, according to the original distribution. The dataset overview is shown in Table 1.

The CityPersons dataset offers significant advantages over existing pedestrian detection datasets. The images in this dataset are collected from multiple cities and represent diverse climatic conditions, offering a wider range of pedestrian types and increasing scene diversity. These features make the CityPersons dataset especially suitable for evaluating the performance of pedestrian detection models in complex urban environments.

4.2 Evaluation Metrics

This study utilizes precision (P), recall (R), mean average precision (mAP), parameters (Params), Giga floating point operations (GFLOPs), and frames per second (FPS) as metrics to assess the performance of our proposed algorithm and other models.

4.2.1 P,R,AP, and mAP

Precision quantifies the ratio of true positives to all predicted positives, whereas recall evaluates the ratio of true positives accurately identified by the model. The equations for accuracy and recall are

$$Precision = \frac{TP}{TP + FP} \tag{11}$$

$$Recall = \frac{TP}{TP + FN}. \tag{12}$$

The AP is calculated as the area beneath the precision-recall curve for a specific class. The mAP is calculated by averaging the AP values for every category. The definitions of AP and mAP are as follows:

Table 2 Experimental settings

Types	Value
Operation System	Ubuntu−20.04
GPU	NVIDIA GeForce RTX 4090
Framework	Pytorch 1.12.1
CUDA	11.3
CuDNN	8.4.0

$$AP = \int_0^1 P_i(R_i) dR_i \tag{13}$$

$$mAP = \frac{\sum_{i=1}^{N} AP_i}{N} \tag{14}$$

where N denotes the aggregate number of categories. mAP is a critical measure for assessing the performance of object detection algorithms.

4.2.2 Parameter Metrics

The metrics of Parameter (Params) denote the quantity of parameters the model acquires throughout the training process. A greater Params number signifies a more intricate model, hence augmenting memory and computing resource demands during training and inference.

4.2.3 GFLOPs

GFLOPs denote the quantity of floating-point operations executed by the model. Similar to Params, GFLOPs quantify the complexity and magnitude of a deep learning model. Higher GFLOPs values need greater processing resources.

4.2.4 FPS

FPS is a crucial metric for assessing the operational speed of an object detection algorithm. It denotes the quantity of images that can be processed per second. Hence, a greater FPS value signifies a faster model, rendering it appropriate for real-time detection tasks.

4.3 Implementation Details

The experimental environment and hardware setups of this study are detailed in Table 2. The initial learning rate is set to 0.001, with a minimum threshold of 0.0001. The input image size is 640 × 640 pixels. The batch size is configured at 8, and all models undergo training for 200 epochs. Furthermore, all other hyperparameters remain at their default configurations.

4.4 Experimental Results

The convergence of the RT-DETR and RT-DETR-improved models across several evaluation metrics are depicted in Figs. 5 and 6. The RT-DETR-improved model exhibits a more rapid and smoother decline in the training set loss curve, signifying accelerated acquisition of feature information throughout the training process. The validation set

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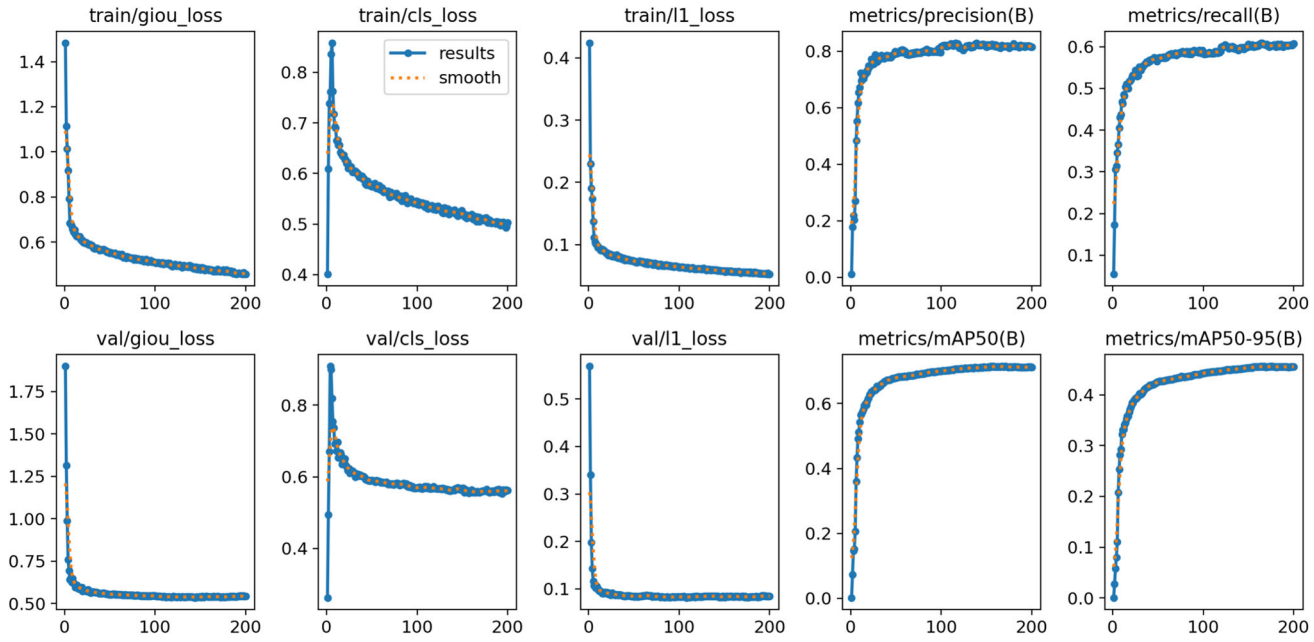


Fig. 5 The convergence curves of RT-DETR on CityPersons dataset

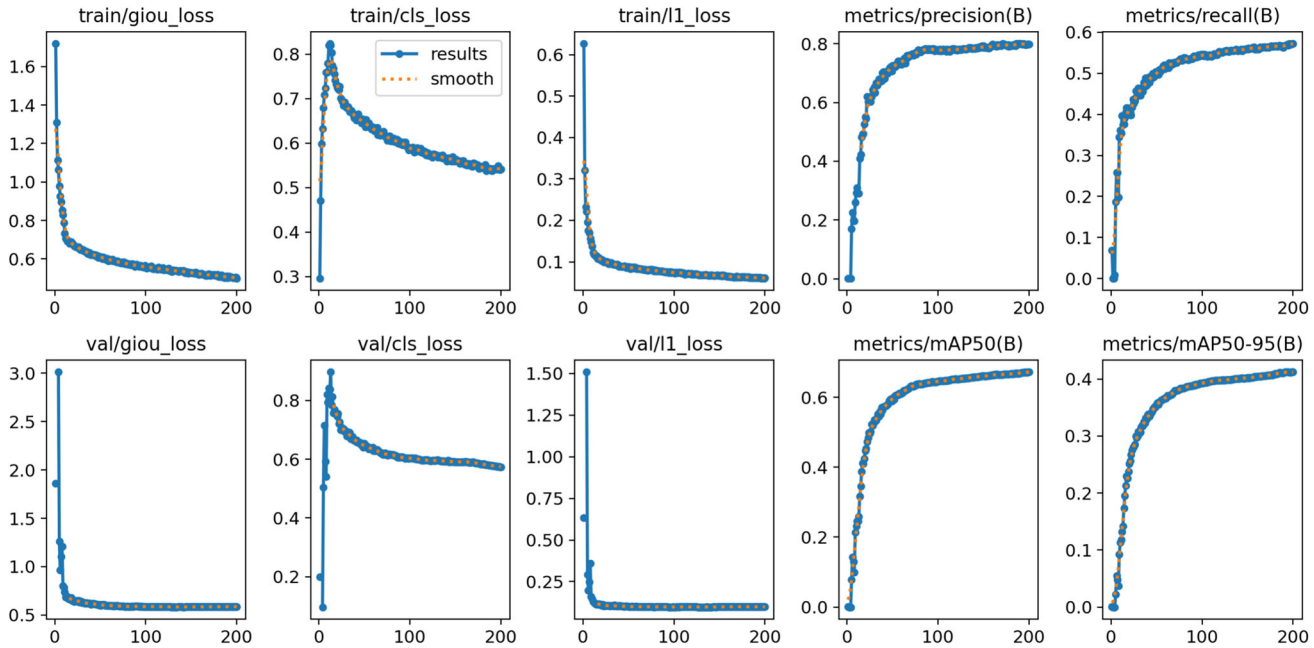


Fig. 6 The convergence curves of RT-DETR-improved on CityPersons dataset

loss curve has a more pronounced drop and reduced loss values, indicating superior prediction accuracy on new data. Furthermore, the accuracy and recall convergence curves indicate that the RT-DETR-improved model converges more rapidly and demonstrates significant enhancements, hence improving its performance in reliably detecting pedestrians. This also minimizes false detection and omissions. The RT-DETR-improved model demonstrates superior performance in mAP@0.5 and mAP@0.5:0.95, especially in consistency and accuracy across various IoU thresholds.

The two confusion matrices generated by the RT-DETR and RT-DETR-improved models on the CityPersons dataset are illustrated in Fig. 7. These matrices demonstrate the predictive performance of each model across all

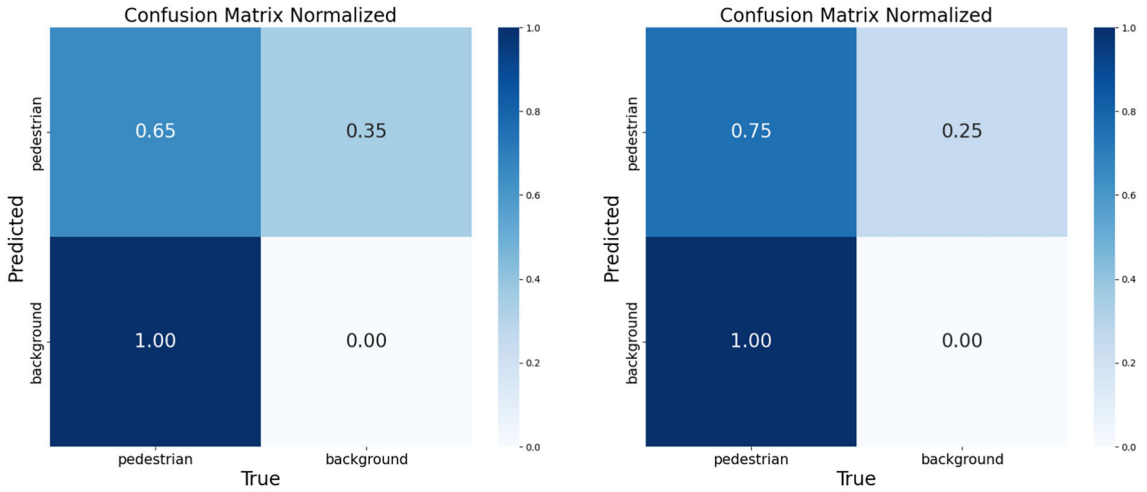


Fig. 7 Comparison of confusion matrices between RT-DETR (left) and RT-DETR-improved (right)

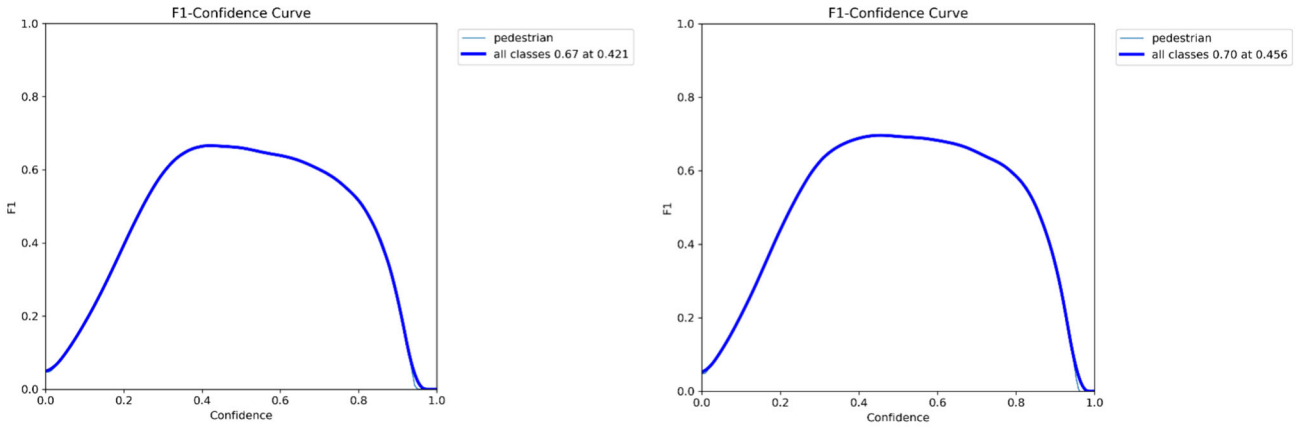


Fig. 8 Comparison of F1-confidence curves between RT-DETR (left) and RT-DETR-improved (right)

categories. The RT-DETR model accurately predicts 65% of pedestrians, while the RT-DETR-improved model accurately predicts 75%. This signifies a 10 percent point enhancement in pedestrian categorization accuracy for the RT-DETR-improved model. Consequently, the RT-DETR-improved model exhibits superior classification capabilities relative to the baseline model.

As shown in Fig. 8, The F1-confidence curves of RT-DETR and RT-DETR-improved illustrate the F1 scores of both models at various confidence thresholds. The RT-DETR model attains its peak F1 score of roughly 0.67 at a confidence level of about 0.421, indicating an improved balance between precision and recall at this threshold. In comparison, the RT-DETR-improved model achieves a superior F1 score of roughly 0.70 at a confidence level of around 0.456. The elevated F1 score and confidence threshold of the RT-DETR-improved model indicate not only enhanced predictive accuracy but also a more efficient distribution of confidence, ensuring precise predictions while minimizing the adverse effects of over- or under-confidence. Thus, the RT-DETR-improved model exhibits distinct advantages in pedestrian identification, especially in contexts necessitating highly reliable predictions.

4.5 Comparative Experiments

To validate the superior performance of the proposed RT-DETR-improved, we conduct a comparative analysis with other leading object detection algorithms, including YOLOv8-X [10] and various end-to-end detection transformers

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Table 3 Comparison results with other models on CityPersons datasets

Model	mAP@0.5	GFLOPs	Params (M)	FPS
YOLOv8-X	71.0	257.8	68.2	56
DETR-DC5	59.0	187	41	45
Anchor-DETR-DC5	67.1	74	16	46
Conditional-DETR	68.7	90	44	48
Efficient-DETR	69.1	159	32	52
DAB-Deformable-DETR	70.2	256	44	53
DN-Deformable-DETR	70.4	195	48	53
DINO-Deformable-DETR	70.7	173	40	55
RT-DETR-improved (ours)	71.4	45	16	62

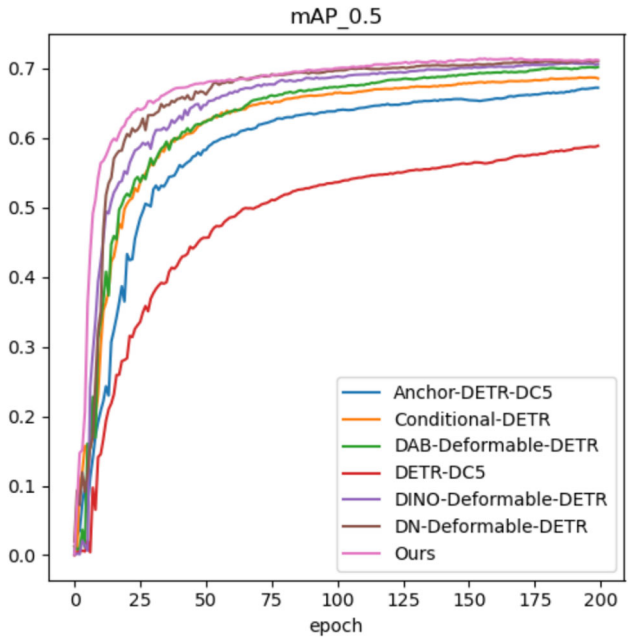


Fig. 9 Comparison curves with other models of mAP@0.5 metric

(DETRs), such as Anchor-DETR-DC5 [37], Conditional-DETR [38], DAB-Deformable-DETR [39], DETR-DC5 [27], DINO-Deformable-DETR [40], and DN-Deformable-DETR [41]. The DETRs utilize ResNet-18 [42] as the backbone network. All models are assessed using the CityPersons dataset, with the results displayed in Table 3.

As shown in Table 3, our proposed RT-DETR-improved model excels in accuracy, computational complexity, parameter count, and inference speed, particularly in terms of FPS and mAP@0.5. Specifically, RT-DETR-improved achieves the highest mAP@0.5 of 71.4, closely followed by YOLOv8-X at 71.0, slightly lower than RT-DETR-improved. However, YOLOv8-X has a computational complexity of 257.8 GFLOPs, the highest among all models, resulting in slower inference speed. Furthermore, our model achieves the best inference speed among all models, demonstrating that it is not only highly accurate but also exceptionally fast, making it ideal for real-time object detection scenarios.

4.6 Ablation Experiments

To verify the performance enhancements of the three optimized components in our proposed methodology, we performed a series of ablation experiments on the CityPersons dataset. The experimental results are presented in Table 4.

Experiment 2 incorporates the HiLo attention mechanism into the AIFI module, enhancing the model's precision by 1.4%, recall by 0.6%, mAP@0.5 by 1.5%, and mAP@0.5:0.95 by 0.3% relative to the baseline model. The results indicate that the HiLo attention mechanism offers superior model representation compared to the traditional

Table 4 The ablation results on CityPersons dataset

Experiment	EAIFI	NFFM	InnerMPDIoU	P (%)	R (%)	mAP@0.5 (%)	mAP@0.5:0.95 (%)
Exp 1				79.1	57.4	67.2	41.2
Exp 2	✓			80.5	58.0	68.7	41.5
Exp 3		✓		78.6	59.3	69.6	42.1
Exp 4	✓	✓		80.7	58.9	70.9	42.8
Exp 5	✓	✓	✓	81.3	60.5	71.4	43.2



Fig. 10 The first set of visual comparison results between RT-DETR (left) and RT-DETR-improved (right) on CityPersons dataset



Fig. 11 The second set of visual comparison results between RT-DETR (left) and RT-DETR-improved (right) on CityPersons dataset

self-attention mechanism. In Experiment 3, our proposed nonlinear feature fusion module achieved a significant 1.9% enhancement in recall. This indicates that the nonlinear feature fusion module proficiently resolves the problem of missed detections, facilitating more comprehensive pedestrian detection. Experiment 4 explored the integration of the HiLo attention mechanism with the nonlinear feature fusion module. The results indicated that the model outperformed each module individually, suggesting that our improvements are compatible and complementary. Finally, the proposed RT-DETR-improved achieves a 2.2% increase in precision, 3.1% in recall, 4.2% in mAP@0.5, and 2.0% in mAP@0.5:0.95, making it more effective and robust for pedestrian detection.

Furthermore, to offer a more intuitive comparison between RT-DETR-improved and the baseline model, the visualization results are presented in Figs. 10, 11, 12. As shown in Fig. 10, the two pedestrians are heavily occluded from each other. The RT-DETR model detects only one pedestrian, while the RT-DETR-improved successfully detects both. As shown in Fig. 11, the RT-DETR-improved detects more small targets, including pedestrians occluded by vehicles. It also successfully detects the irregularly-shaped pedestrian on the far right, wearing a hat, which the RT-DETR fails to identify. As shown in Fig. 12, the RT-DETR model suffers from false detections, such as misidentifying a bicycle behind the black car on the left as a pedestrian. In contrast, the improved model does not generate false detections and can detect pedestrians occluded by large trees or crouching, which the RT-



Fig. 12 The third set of visual comparison results between RT-DETR (left) and RT-DETR-improved (right) on CityPersons dataset

DETR model fails to identify. The RT-DETR-improved model demonstrates superior anti-interference capabilities, particularly in handling intricate situations such as pedestrian position variability and occlusion. Compared to the baseline, RT-DETR-improved significantly reduces the occurrences of missed and false detections, thereby effectively improving the model's overall detection performance.

5 Conclusion

In this study, we propose the RT-DETR-improved model for pedestrian detection. First, we incorporate the HiLo attention mechanism to augment the AIFI module. Furthermore, we present a nonlinear feature fusion module together with the InnerMPDIoU loss function. Compared to the baseline model, RT-DETR-improved significantly boosts mAP@0.5 by 4.2%, mAP@0.5:0.95 by 2.0%, precision by 2.2%, and recall by 3.1% on the CityPersons dataset. RT-DETR-improved exhibits superior effectiveness in pedestrian detection, particularly in overcoming issues such as missed detections, false detections, and inaccurate prediction boxes, making it more reliable in intricate urban settings.

In further research, we will concentrate on optimizing the proposed model to provide fast, real-time pedestrian detection, particularly in environments with constrained computational resources. Furthermore, we will refine the model architecture and augment its capacity to adjust to complex situations, hence enhancing its generalization efficacy.

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Declarations

Conflict of Interest The authors declare no conflict of interest.

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